

How does Locality and Unitarity constrain the structure of a CFT with boundary?

May 11, 2026

Abstract. This is a note capturing the main idea and calculation of [1], which focus on how sewing manifold will constraint the sturcture of a CFT with boundary presented. For A series minimal model a set of solutions in a special normalization is given by [2].

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1 Bulk Sewing Constraint

1.1 Bulk Content

1.1.1 Data Convention

Here we set up the convention a Bulk CFT data. The correlation function of a bulk CFT can be defined with the following data:

- **Lower Point Function** Convention:

$$\begin{aligned}\langle\phi_i\rangle &= 0, & \langle I\rangle &= 1, \\ \langle\phi_i(z,\bar{z})\phi_j(0,0)\rangle &= \frac{\delta_{ij}}{(z\bar{z})^{2\Delta_i}},\end{aligned}\tag{1}$$

We take this convention of bulk 2 point function is always normalized. Then we define the bulk OPE coefficient C_{ij}^k as:

- **Bulk Field OPE** Coefficient Convention:

$$\phi_i(z,\bar{z})\phi_j(0,0) = \sum_k C_{ij}^k \phi_k z^{\Delta_k-\Delta_i-\Delta_j} \bar{z}^{\Delta_k-\Delta_i-\Delta_j} + \text{Descendants...}\tag{2}$$

Then with this convention we can talk about the constraints these data should satisfy due to the consistency of the CFT.

There are 2 Bulk Sewing Constraints for a CFT. The first one is the crossing symmetry of the 4 point function. The second one is the modular invariance of the torus partition function. Now let's talk about the first one.

1.2 Crossing Symmetry

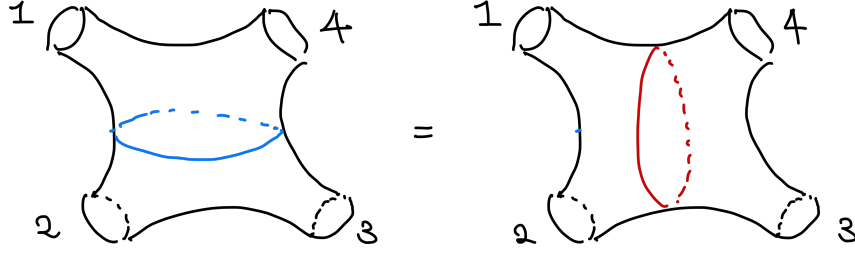


Figure 1: Crossing Symmetry of 4 point function. The 4 point function can be decomposed into 3 different channels, and the crossing symmetry means that these 3 different decompositions should be the same function.

1.2.1 4 point function Decomposition

The 4 point function of a CFT can be written in the following way.

- First, the global Ward Identity tells us that the 4 point function can be written as some function of the cross ratio:

$$\langle \phi_1(z_1)\phi_3(z_2)\phi_3(z_3)\phi_4(z_4) \rangle = \prod_{i<j} (z_i - z_j)^{r-\Delta_i-\Delta_j} (\bar{z}_i - \bar{z}_j)^{\bar{r}-\bar{\Delta}_i-\bar{\Delta}_j} Y(\eta, \bar{\eta}), \quad (3)$$

where

$$r = \frac{1}{3} \left(\sum_i \Delta_i \right) \quad \bar{r} = \frac{1}{3} \left(\sum_i \bar{\Delta}_i \right) \quad (4)$$

and

$$\eta = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_3)(z_2 - z_4)} \quad (5)$$

is the cross ratio.

- Second, the OPE and Local Ward Identity gives us the following expressions of $Y(\eta, \bar{\eta})$:

$$\begin{aligned} Y(\eta, \bar{\eta}) &= \sum_k C_{12}^k C_{34}^k \mathcal{F}(12, 34; k)(\eta) \bar{\mathcal{F}}(12, 34; k)(\bar{\eta}) \\ &= \sum_k C_{14}^k C_{23}^k \mathcal{F}(14, 23; k)(1-\eta) \bar{\mathcal{F}}(14, 23; k)(1-\bar{\eta}) \\ &= \sum_k C_{13}^k C_{24}^k \mathcal{F}(13, 24; k) \left(\frac{1}{\eta} \right) \bar{\mathcal{F}}(13, 24; k) \left(\frac{1}{\bar{\eta}} \right) \end{aligned} \quad (6)$$

notice that:

$$\eta = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_3)(z_2 - z_4)} \quad 1 - \eta = \frac{(z_1 - z_4)(z_2 - z_3)}{(z_1 - z_3)(z_2 - z_4)} \quad \frac{1}{\eta} = \frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_2)(z_3 - z_4)} \quad (7)$$

which is the cross ratio of the 4 points in different channels.

1.2.2 Conformal Blocks

The special function above we use: $\mathcal{F}(12, 34; k)(\eta)$ is called the conformal block. It is a special function determined by the conformal symmetry. Here let's list out some basic properties of the conformal block:

1. These functions are **Chiral** and only depends on Chiral data. Thus, they are multivalued.
2. 4 point Conformal Blocks on a plane are multivalued and have branch cuts. The cutting points are at $\eta = 0, 1, \infty$. The conformal blocks are analytic on the complex plane with these cuts.

Remark

Indeed there are some subtleties about the convention, though we write i, j, k, l in the convention, it in fact only depends on the chiral data of the double fold virasoro primary.

The most important fact of the conformal blocks is that they satisfy some linear relations called duality relations. For 4 point blocks on a plane, its famous for the duality relation called the **F-move**:

$$\mathcal{F}(ij, kl; p)(\eta) = \sum_q F_{pq} \begin{pmatrix} i & l \\ j & k \end{pmatrix} \mathcal{F}(il, jk; q)(1 - \eta) \quad (8)$$

here $F_{pq} \begin{pmatrix} i & j \\ k & l \end{pmatrix}$ is called the F-matrix, which only depends on the chiral data of the CFT. They satisfies some consistency conditions which can eventually be determined analytically.

Note

In fact this duality relation of conformal blocks exactly follows from the expectation of the crossing symmetry to hold, if we assume finite primary fields .

1.2.3 Crossing Symmetry

The crossing symmetry is the following statement:

- The 4 point function should be single valued and analytic on the whole complex plane. This means that the 3 different conformal blocks decomposition of the 4 point function should be the same function.

This statement can be written in a mathematical way as:

$$\begin{aligned} & \sum_p C_{12}^p C_{34}^p \mathcal{F}(12, 34; p)(\eta) \overline{\mathcal{F}}(12, 34; p)(\bar{\eta}) \\ &= \sum_q C_{14}^q C_{23}^q \mathcal{F}(14, 23; q)(1 - \eta) \overline{\mathcal{F}}(14, 23; q)(1 - \bar{\eta}) \end{aligned} \quad (9)$$

if we plug in the F-move, we can get:

Theorem 1.1

Crossing Symmetry

$$\sum_p C_{12}^p C_{34}^p F_{pq} \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} \overline{F}_{pl} \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} = C_{14}^q C_{23}^q \delta_{ql} \quad (10)$$

notice that the consistency condition of conformal blocks makes the F-matrix satisfy the following normalization condition:

$$\sum_q F_{pq} \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} \overline{F}_{ql} \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} = \delta_{pl} \quad (11)$$

Remark

Notice that $\bar{F}$ is the complex conjugate of F . In some cases the F-matrix is real, then we can just write F instead of $\bar{F}$.

Thus the crossing symmetry can be written in a more compact way as:

$$C_{12}^p C_{34}^p F_{pr} \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} = C_{14}^r C_{23}^r F_{rp} \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} \quad (12)$$

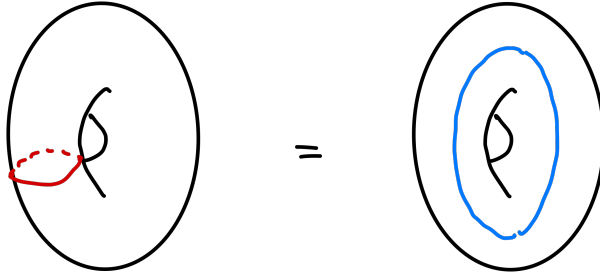
1.3 Modular Invariance

Figure 2: Modular S Invariance of Torus Partition Function. The torus partition function should be invariant under the modular S transformation.

1.3.1 Modular S Invariance

Modular invariant is that the Torus partition function should be invariant under modular transformation. Particularly, we care most about the modular S transformation. The torus partition function invariant under the modular S transformation means that the partition function should satisfy the following condition:

$$Z(\tau) = Z\left(-\frac{1}{\tau}\right) \quad (13)$$

Torus partition function can be decomposed into characters:

$$Z(\tau) = \sum_{i,j} \chi_p(\tau) \bar{\chi}_p(\bar{\tau}) \quad (14)$$

Thus modular S invariance means that the characters should satisfy the following condition:

Theorem 1.2**Modular S Invariance**

$$\sum_p \chi_p(\tau) \bar{\chi}_p(\bar{\tau}) = \sum_p \chi_p\left(-\frac{1}{\tau}\right) \bar{\chi}_p\left(-\frac{1}{\bar{\tau}}\right) \quad (15)$$

1.3.2 Character S Transformation

Just as the 4 point blocks transform under the F move in different channels. The CFT characters have following behavior:

$$\chi_i(\tau) = \sum_j S_{ij} \chi_j\left(-\frac{1}{\tau}\right) \quad (16)$$

where S_{ij} is called the S-matrix, which only depends on the chiral data of the CFT. For RCFT or CFT corresponding to a Modular Tensor Category. The S-matrix is symmetric.

Note

In fact the S transformation rule have this form directly follows from expecting the torus partition function to be modular invariant.

2 Boundary Sewing Constraint

Now we admit the CFT to have a boundary on the real axis. Now we to restric the conformal symmetry to preserve the boundary, which lead to the **conformal boundary condition**;

$$T(z) = \overline{T}(\bar{z}) \quad z = \bar{z} \quad (17)$$

with this boundary condition, the theory here only have 1 fold of Virasoro symmetry.

2.1 Boundary Content

2.1.1 Theory Contents

With the boundary presented, we need more data to characterize the CFT.

1. Boundary Conditions a
2. Boundary Operators $\psi_i^{(ab)}$, here i is the label of 1 fold Virasoro primary operator, and a, b are the labels of the boundary conditions.

2.1.2 Data Convention

A full CFT with boundary on UHP can be characterized by the following data:

- **Lower Point Function** Convention:

$$\langle \psi_i^{(ab)} \rangle = 0, \quad \langle I \rangle_a = \alpha^a \quad (18)$$

Notice that here, with the boundary presented and with the bulk convention fixed, we no longer can normalized the boundary 0 point function freely.

- **Boundary 2 Point Function** Convention:

$$\langle \psi_i^{(ab)}(0) \psi_j^{(cd)}(x) \rangle = \delta_{ij} \delta_{ad} \delta_{bc} x^{-2\Delta_i} \alpha_i^{(ab)}, \quad x > 0 \quad (19)$$

¹ Indeed, we can choose to normalize the boundary field itself to make the boundary 2 point function to be normalized. However, this won't be a natural choice.

- **Boundary OPE** Coefficient Convention:

$$\psi_i^{(ab)}(0) \psi_j^{(bc)}(x) = \sum_k C_{ij}^{(abc)k} x^{\Delta_k - \Delta_i - \Delta_j} \psi_k^{(ac)}(0) + \text{descendants} \dots \quad (20)$$

- **Bulk-Boundary OPE** Coefficient Convention:

$$\phi_i(z, \bar{z}) = \sum_k B_i^{(a)k} (2y)^{\Delta_k - \Delta_i - \bar{\Delta}_i} \psi_k^{(aa)}(x) + \text{descendants} \dots \quad (21)$$

Here the bulk field only fuse to boundary field with the same boundary condition. The factor 2 is from the folding trick.

2.1.3 Symmetry of OPE Coefficient

We can conformally map the UHP to a unit disk, then we can see that the two point structure constant and the OPE coefficient should satisfy the following symmetry:

- **Symmetry of 2 Point Structure Constant**, due to the fact of symmetry:

¹The notaion here is different from Lewellen's paper, I guess he had a typo in the original paper.

$$\alpha_i^{(ab)} = \alpha_i^{(ba)} \quad (22)$$

- **Symmetry of OPE Coefficient**, due to the fact of symmetry of 3 point structural constant:

$$C_{ijk}^{(abc)} = C_{jki}^{(bca)} = C_{kij}^{(cab)} \quad (23)$$

we know that the OPE coefficients are related to the 3 point structure constant by the following relation:

$$C_{ij}^{(abc)k} \alpha_k^{(ac)} = C_{ijk}^{(abc)} \quad (24)$$

Thus we have a symmetry of OPE coefficient:

$$C_{ij}^{(abc)k} \alpha_k^{(ac)} = C_{jk}^{(bca)i} \alpha_i^{(ba)} = C_{ki}^{(cab)j} \alpha_j^{(cb)} \quad (25)$$

2.2 Sewing Constraints

2.2.1 Cardy's Condition

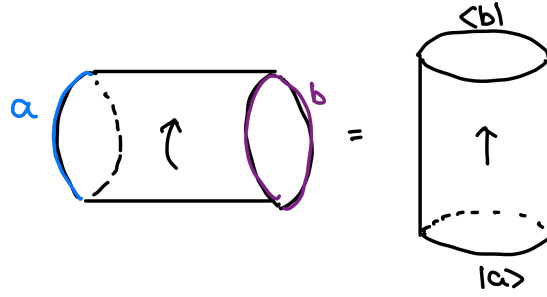


Figure 3: Cardy's condition of possible boundary conditions.

The first constraint is the Cardy's condition of possible boundary conditions:

Theorem 2.1

Cardy's Condition

$$\begin{aligned} Z_{ab}(\tau) &= \text{Tr}_{ab} e^{2\pi i \tau (L_0 - c/24)} = \sum_i N_i^{ab} \chi_i(\tau) \\ &= \langle a | e^{-i\pi/\tau (L_0 + \bar{L}_0 - c/12)} | b \rangle. \end{aligned} \quad (26)$$

Here $|a\rangle, |b\rangle$ are boundary states corresponding to the boundary conditions a, b .

2.2.1.1 Ishibashi State

On the Cylinder the conformal boundary condition can be written as:

$$(L_n - \bar{L}_{-n})|B\rangle = 0 \quad (27)$$

A solution to this equation is the Ishibashi state:

$$|i\rangle = \sum_N |i, N\rangle \otimes |\bar{i}, N\rangle \quad (28)$$

notice that the i primary should be diagonal $\Delta_i = \bar{\Delta}_i$.

2.2.1.2 Solution of Cardy's Condition

A series of solution for the Cardy's condition is given by the following formula for **Diagonal RCFT**, this is discussed in [3]:

$$|a\rangle = \sum_i \left(\frac{S_{ai}}{\sqrt{S_{0i}}} \right) |i\rangle \quad (29)$$

Thus the boundary conditions here are labeled by the bulk primary field. This lead to a idea of:

- For a boundary field $\psi_i^{(ab)}$ to exist, it need to obey the bulk fusion rule: $N_{ab}^i \neq 0^2$

This is an observation for diagonal RCFT, however, for general CFT this may not work and need to be modified.

2.2.1.3 0 point function and Bulk Boundary OPE

The solution Equation 29 also give us a nontrivial 0 point function and the bulk boundary OPE coefficient for identity boundary fields. These statements are discussed in [4]. Here we just list a consistent result:

- **0 point function**

$$\langle I \rangle_a = \alpha^a = \frac{S_{a0}}{\sqrt{S_{00}}} \quad (30)$$

- **Bulk Boundary OPE Coefficient** with boundary identity field:

$$B_i^{(a)0} = \frac{S_{ai}}{S_{a0}} \left(\frac{S_{00}}{S_{0i}} \right)^{1/2} \quad (31)$$

2.2.2 Crossing Symmetry

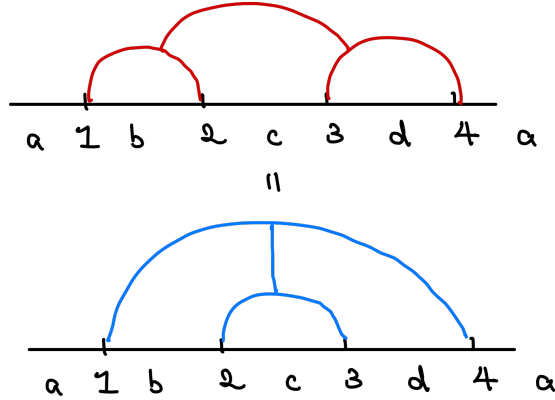


Figure 4: Crossing Symmetry of 4 point function with boundary.

2.2.2.1 Sewing Analysis

The boundary 4 point function can be written as:

$$\langle \psi_1^{(ab)}(x_1) \psi_2^{(bc)}(x_2) \psi_3^{(cd)}(x_3) \psi_4^{(da)}(x_4) \rangle = \prod_{i < j} (x_i - x_j)^{r - \Delta_i - \Delta_j} Y(\eta), \quad (32)$$

$$x_4 > x_3 > x_2 > x_1,$$

and the cross ratio is defined as:

$$\eta = \frac{(x_1 - x_2)(x_3 - x_4)}{(x_1 - x_3)(x_2 - x_4)} \quad (33)$$

²I may take this statement more as a conjecture of a possible structure.

Similarly for the bulk boundary 4 point function, we can also decompose it into two different channels, the s channel and the t channel:

$$\begin{aligned} Y(\eta) &= \sum_k C_{12}^{(abc)k} C_{34}^{(cda)k} \alpha_p^{(ac)} \mathcal{F}(12, 34; k)(\eta) \\ &= \sum_k C_{23}^{(bcd)k} C_{41}^{(dab)k} \alpha_k^{(bd)} \mathcal{F}(14, 23; k)(1 - \eta) \end{aligned} \quad (34)$$

Remark

Multi Valuedness of Conformal Block

Notice that the conformal block here is multi-valued, and we need to specify the branch cut to make it single valued. The above function we chose a branch cut for η from 0 to ∞ and 1 to ∞ . Thus the correlation function with $x_4 > x_3 > x_2 > x_1$ is single valued.

2.2.2.2 Boundary Crossing Symmetry

Same as the bulk crossing symmetry, the boundary crossing symmetry also give us a constraint on the structure constant:

Theorem 2.2

Boundary Crossing Symmetry

$$\sum_k C_{12}^{(abc)k} C_{34}^{(cda)k} \alpha_k^{(ac)} F_{kp} \left(\begin{matrix} 1 & 4 \\ 2 & 3 \end{matrix} \right) = C_{41}^{(dab)p} C_{23}^{(bcd)p} \alpha_p^{(bd)} \quad (35)$$

2.2.3 Boundary-Boundary-Bulk Sewing Constraint

Consider a Bulk one point and boundary two point configuration.

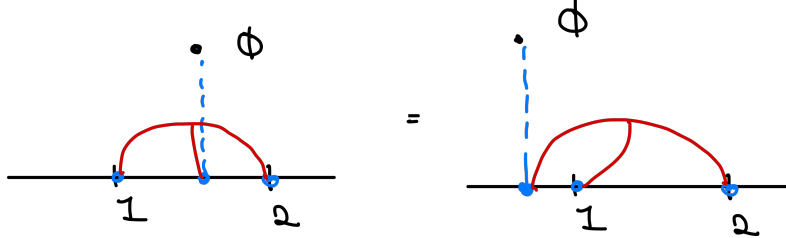


Figure 5: Bulk-Boundary-Boundary Sewing Constraint.

2.2.3.1 Sewing Analysis

The fields are labeled by:

$$\phi_i(z, \bar{z}) \quad \psi_1^{(ab)}(x_1) \quad \psi_2^{(ba)}(x_2) \quad x_2 > x_1 > 0 \quad (36)$$

However, the difference of the LHS and RHS is that:

- LHS we have

$$x_1 < \text{Re}(z) < x_2 \quad (37)$$

- RHS we have

$$\text{Re}(z) < x_1 < x_2 \quad (38)$$

The two configuration can be equal in the sense of analytical continuing the LHS picture form $\text{Re}(z) > x_1$ to $\text{Re}(z) < x_1$.

On both side we can write the correlation function in terms of the conformal blocks with the cross ratio:

$$\eta \equiv \frac{(z - \bar{z})(x_1 - x_2)}{(z - x_1)(\bar{z} - x_2)}. \quad (39)$$

As usual, the conformal blocks is multi-valued and needs a branch cut. We choose the branch cut of:

$$\mathcal{F}(i\bar{i}, 12; k)(\eta) \quad (40)$$

at from $-\infty$ to 1 to ensure that the correlation function is single valued as $x_2 > x_1$ and z is in the upper half plane.

However, as we assume that ϕ field approaches the boundary, the two cases beahaves differently:

- LHS: as ϕ approaches the boundary, the cross ratio η approaches 0 from the UHP.
- RHS: as ϕ approaches the boundary, the cross ratio η approaches 0 from the LHP.

This discussion is just of a rough picture. For detailed analysis see [1].

2.2.3.2 Boundary-Boundary-Bulk Sewing Constraint

Now we analytical continue the LHS picture from η approaches 0 from UHP to η approaches 0 from LHP. We do this by turning clockwise around the branch cut.

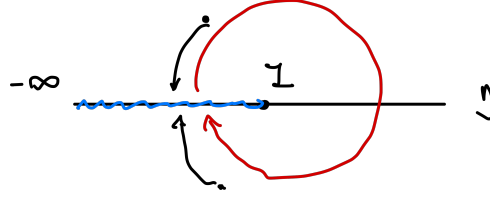


Figure 6: Analytical continuation of the Bulk-Boundary-Boundary configuration.

Under this continuation, in fact we can express in terms of the duality matrix and give out a sewing constraint:

Theorem 2.3

Boundary-Boundary-Bulk Sewing Constraint

$$\begin{aligned} & \sum_p \sum_s B_i^{(b)p} C_{21}^{(bab)p} \alpha_p^{(bb)} F_{ps} \begin{pmatrix} i & 2 \\ \bar{i} & 1 \end{pmatrix} F_{sq} \begin{pmatrix} i & \bar{i} \\ 2 & 1 \end{pmatrix} e^{i\pi(\Delta_1 + \Delta_2 + 2\Delta_i - 2\Delta_s - 2\Delta_q + \Delta_q)} \\ &= B_i^{(a)q} C_{12}^{(aba)q} \alpha_q^{(aa)} \end{aligned} \quad (41)$$

2.2.4 Bulk-Bulk Sewing Constraint

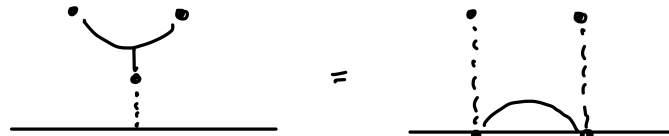


Figure 7: Bulk-Bulk Sewing Constraint.

2.2.4.1 Sewing Analysis

This sewing constraint is about 2 bulk fields configuration. There are two different ways to fuse the two bulk fields:

1. Fuse two bulk field into 1 and then calculate the 1 point function of the bulk field.
2. Fuse each bulk field into boundary field and then calculate the 2 point function of the boundary field.

2.2.4.2 Bulk-Bulk Sewing Constraint

This gives out a constraint on the bulk-boundary OPE coefficient, given bulk OPE and boundary 2 point function:

Theorem 2.4

Bulk-Bulk Sewing Constraint

$$B_i^{(a)q} B_j^{(a)q} \alpha_q^{(aa)} = \sum_m C_{ji}^m B_m^{(a)I} \alpha^a F_{mq} \begin{pmatrix} i & \bar{i} \\ j & \bar{j} \end{pmatrix} e^{(i\pi/2)[\Delta_j - \bar{\Delta}_j + \bar{\Delta}_i - \Delta_i]} \quad (42)$$

2.2.5 Bulk-Bulk-Boundary Sewing Constraint

The final one is the bulk-bulk-boundary sewing constraint:

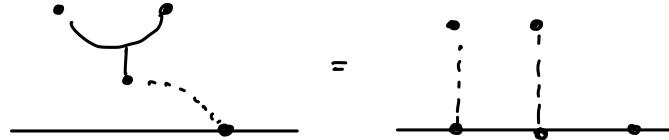


Figure 8: Bulk-Bulk-Boundary Sewing Constraint.

This constraint need a more detailed analysis of analytical structure and constains analysis of 5 point conformal block, thus we won't give the detailed analysis here.

The result is:

Theorem 2.5

Bulk-Bulk-Boundary Sewing Constraint

$$\begin{aligned} & C_{tq}^{(aaa)\psi} B_1^{(a)t} B_2^{(a)q} \\ &= \sum_p \sum_r C_{21}^p B_p^{(a)\psi} F_{pr} \begin{pmatrix} 1 & \psi \\ 2 & \bar{p} \end{pmatrix} F_{\bar{p},q} \begin{pmatrix} 2 & \bar{2} \\ r & \bar{1} \end{pmatrix} F_{rt} \begin{pmatrix} 1 & \bar{1} \\ \psi & q \end{pmatrix} \\ & \times e^{(i\frac{\pi}{2})(\Delta_\psi + \Delta_1 + \bar{\Delta}_1 + \Delta_2 - \bar{\Delta}_2 - \Delta_p + \bar{\Delta}_p - 2\Delta_r + \Delta_q - \Delta_t)} \end{aligned} \quad (43)$$

2.3 Summary

Here I list out all sewing constraints with boundary presented:

- Boundary Crossing Symmetry:

$$\sum_k C_{12}^{(abc)k} C_{34}^{(cda)k} \alpha_k^{(ac)} F_{kp} \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} = C_{41}^{(dab)p} C_{23}^{(bcd)p} \alpha_p^{(bd)} \quad (44)$$

- Boundary-Boundary-Bulk Sewing Constraint:

$$\begin{aligned} & \sum_p \sum_s B_i^{(b)p} C_{21}^{(bab)p} \alpha_p^{(bb)} F_{ps} \begin{pmatrix} i & 2 \\ \bar{i} & 1 \end{pmatrix} F_{sq} \begin{pmatrix} i & \bar{i} \\ 2 & 1 \end{pmatrix} e^{i\pi(\Delta_1+\Delta_2+2\Delta_i-2\Delta_s-2\Delta_q+\Delta_q)} \\ & = B_i^{(a)q} C_{12}^{(aba)q} \alpha_q^{(aa)} \end{aligned} \quad (45)$$

- Bulk-Bulk Sewing Constraint:

$$B_i^{(a)q} B_j^{(a)q} \alpha_q^{(aa)} = \sum_m C_{ji}^m B_m^{(a)I} \alpha^a F_{mq} \begin{pmatrix} i & \bar{i} \\ j & \bar{j} \end{pmatrix} e^{(i\pi/2)[\Delta_j-\bar{\Delta}_j+\bar{\Delta}_i-\Delta_i]} \quad (46)$$

- Bulk-Bulk-Boundary Sewing Constraint:

$$\begin{aligned} & C_{tq}^{(aaa)\psi} B_1^{(a)t} B_2^{(a)q} \\ & = \sum_p \sum_r C_{21}^p B_p^{(a)\psi} F_{pr} \begin{pmatrix} 1 & \psi \\ 2 & \bar{p} \end{pmatrix} F_{\bar{p},q} \begin{pmatrix} 2 & \bar{2} \\ r & \bar{1} \end{pmatrix} F_{rt} \begin{pmatrix} 1 & \bar{1} \\ \psi & q \end{pmatrix} \\ & \times e^{(i\frac{\pi}{2})(\Delta_\psi+\Delta_1+\bar{\Delta}_1+\Delta_2-\bar{\Delta}_2-\Delta_p+\bar{\Delta}_p-2\Delta_r+\Delta_q-\Delta_t)} \end{aligned} \quad (47)$$

3 Example: Ising CFT

3.1 Basic Data

This subsection we list out some basic datas of Ising CFT, within convention of [1].

3.1.1 Field and Fusion Rule

The Ising CFT has 3 primary fields, which are the identity field I , the spin field σ and the energy field ε . The fusion rule of these fields are:

$$\begin{aligned} I \times I &= I, & I \times \sigma &= \sigma, & I \times \varepsilon &= \varepsilon, \\ \sigma \times \sigma &= I + \varepsilon, & \sigma \times \varepsilon &= \sigma, & \varepsilon \times \varepsilon &= I \end{aligned} \quad (48)$$

Thus the only nontrivial bulk 4 point functions are:

$$\begin{aligned} &\langle \sigma(z_1, \bar{z}_1) \sigma(z_2, \bar{z}_2) \sigma(z_3, \bar{z}_3) \sigma(z_4, \bar{z}_4) \rangle, \\ &\langle \varepsilon(z_1, \bar{z}_1) \varepsilon(z_2, \bar{z}_2) \varepsilon(z_3, \bar{z}_3) \varepsilon(z_4, \bar{z}_4) \rangle \\ &\langle \varepsilon(z_1, \bar{z}_1) \varepsilon(z_2, \bar{z}_2) \sigma(z_3, \bar{z}_3) \sigma(z_4, \bar{z}_4) \rangle \end{aligned} \quad (49)$$

3.1.2 F Matrix and S Matrix

The F Matrix of Ising CFT is given by:

$$\begin{aligned} F_{00} \begin{pmatrix} \sigma & \sigma \\ \sigma & \sigma \end{pmatrix} &= \frac{1}{\sqrt{2}}, & F_{0\varepsilon} \begin{pmatrix} \sigma & \sigma \\ \sigma & \sigma \end{pmatrix} &= \frac{1}{2\sqrt{2}}, & F_{\varepsilon 0} \begin{pmatrix} \sigma & \sigma \\ \sigma & \sigma \end{pmatrix} &= \sqrt{2}, & F_{\varepsilon\varepsilon} \begin{pmatrix} \sigma & \sigma \\ \sigma & \sigma \end{pmatrix} &= -\frac{1}{\sqrt{2}}, \\ F_{0\sigma} \begin{pmatrix} \sigma & \varepsilon \\ \sigma & \varepsilon \end{pmatrix} &= \frac{1}{2}, & F_{\sigma 0} \begin{pmatrix} \varepsilon & \varepsilon \\ \sigma & \sigma \end{pmatrix} &= 2, & F_{\sigma\sigma} \begin{pmatrix} \varepsilon & \sigma \\ \sigma & \varepsilon \end{pmatrix} &= -1, & F_{00} \begin{pmatrix} \varepsilon & \varepsilon \\ \varepsilon & \varepsilon \end{pmatrix} &= 1, \end{aligned} \quad (50)$$

The S Matrix of Ising CFT is given by:

$$S = \frac{1}{2} \begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{pmatrix} \quad (51)$$

3.2 Bulk and Boundary Data

Now we try to solve the sewing constraints of Ising CFT.

3.2.1 Basic Data

- **Bulk OPE:** Simply Solving [Theorem 1.1](#) for the 4 point function of Ising CFT:

$$C_{\sigma\sigma}^I = 1, \quad C_{\varepsilon\varepsilon}^I = 1, \quad C_{\sigma\sigma}^\varepsilon = \frac{1}{2}, \quad (52)$$

- **Boundary Conditions** Solving the Cardy's condition [Theorem 2.1](#) we can immediatly find three boundary conditions corresponding to the three primary fields:

$$I \sim |+\rangle \quad \varepsilon \sim |-\rangle \quad \sigma \sim |f\rangle \quad (53)$$

the cardy states can be explicitly written out.

- **Boundary Field Content** due to the claim in Section 2.2.1.2, the boundary operator content can be read out from the fusion rule, which is:

	+	−	f
+	$I^{(++)}$	$\psi^{(+-)}$	$\sigma^{(+f)}$
−	$\psi^{(-+)}$	$I^{(--)}$	$\sigma^{(-f)}$
f	$\sigma^{(f+)}$	$\sigma^{(f-)}$	$I^{(ff)}, \psi^{(ff)}$

Table 1: Boundary Operator of Ising

3.2.2 Boundary Sewing Constraint

A trivial one is the Boundary 0 point function and bulk-boundary I OPE, which follows directly from Cardy's condition.

- **Boundary 0 Point** the result Equation 30 for the 0 point function normalization is:

$$\alpha^+ = \alpha^- = \frac{1}{\sqrt{2}}, \quad \alpha^f = 1 \quad (54)$$

- **Bulk-Boundary I OPE** for OPE with identity field, we follow from Equation 31 to get:

$$\begin{aligned} B_\epsilon^{(\pm)I} &= 1 & B_\epsilon^{(f)I} &= -1 \\ B_\sigma^{(f)I} &= 0 & B_\sigma^{(\pm)\sigma} &= \pm 2^{1/4} \end{aligned} \quad (55)$$

4 Runkel's solution for A series minimal model

In the paper [2], Runkel gives a full solution to the sewing constraint for A series minimal model. The solution is given in a special normalization, which makes the result simple and elegant.

4.1 BCO Sewing

The BCO OPE can be fixed up to some freedom with the boundary crossing equation itself.

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